Visualising the Effect of Kernel Functions on Decision Boundaries in Support Vector Machines

MODULE: MACHINE LEARNING AND NEURAL NETWORKS
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ABSTRACT:

Support Vector Machines (SVMs) are a powerful supervised learning algorithm in machine learning, widely used for classification tasks. They are mainly successful in high-dimensional spaces and can model both linear and non-linear relationships using kernel functions.
One of the key features of SVMs is the use of kernel functions, which allow the algorithm to model non-linear decision boundaries. This report investigates the effect of different kernel functions on the behaviour and performance of SVM classifiers using a synthetic dataset. Three kernels are examined: linear, polynomial, and radial basis function (RBF).

A series of visualisations demonstrate how kernel choice affects decision boundary shape and classification performance. In addition, a quantitative comparison and hyperparameter analysis are conducted to evaluate model behaviour. The results show that non-linear kernels significantly improve classification performance for datasets with complex structures. In particular, the RBF kernel produces the most flexible decision boundary and achieves the highest accuracy.

INTRODUCTION:

Machine learning programs support decisions in areas like finance, healthcare, and computer vision also to find patterns in complicated data and support. In machine learning classification programs sort things into set groups based on their features. Support Vector Machines (SVMs) are known for their reliable methods and accurate predictions.

Support Vector Machines perform by recognizing an optimal decision boundary that separates different classes within a dataset. This algorithm constructs a hyperplane that maximises the margin between classes, which improves the model's ability to generalise to unseen data according to Cortes and Vapnik, 1995. While linear SVM models can perform well when the data is linearly separable, many real-world datasets exhibit complex patterns that cannot be separated by a simple straight line.

Kernel functions require a powerful mechanism for extending SVM models to non-linear problems. By mapping data into higher dimensional feature spaces, this is explained by Bishop, 2006. This technique is commonly referred to as the **kernel trick.**

Three kernel functions were examined in this study:

1. **Linear Kernel**: The simplest kernel function is linear kernel function and makes a straight decision boundary. It is computationally efficient but limited when the dataset includes complex patterns.

2. **Polynomial Kernel:** The polynomial kernel announces non-linearity by modelling interactions between features. This permits the classifier to create curved decision boundaries that better fit datasets with non-linear structures.

3. **Radial Basis Function Kernel:** This kernel function is called Radical Basis function, is often used in machine learning because it can measure the distance between data points, which helps sort data in separate ways. The RBF kernel uses a setting called gamma, which controls how much each data point affects the model (James et al., 2023).

BACKGROUND OF SVM:

Support Vector Machines (SVMs) are supervised learning algorithms used for classification and regression tasks. The method was first introduced by Cortes and Vapnik (1995), who proposed an algorithm for constructing optimal decision boundaries that maximise the margin between classes. The central idea behind SVM classification is to identify a hyperplane that separates data points belonging to distinct categories while maximising the distance between the boundary and the nearest training examples.

The nearest data points are known as **support vectors and are essential** in determining the position of the decision boundary. Maximising the margin between classes enables support vector machine (SVM) models to enhance generalisation performance and mitigate the risk of overfitting (Cortes and Vapnik, 1995).

To handle such cases, SVMs use kernel methods, which map data into higher-dimensional feature spaces. This process, often referred to as the kernel trick, allows the algorithm to separate data that is not linearly separable in its original form without explicitly performing complex transformations.

Several kernel functions can be used, each offering different ways to model relationships within the data. Common examples include the linear kernel, polynomial kernel, and radial basis function (RBF) kernel. Choosing an appropriate kernel is essential, as it directly affects the model's performance and its ability to capture patterns in the data.

Dataset and Experimental Setup:

To demonstrate the behaviour of different kernel functions, this study uses the make_moons dataset from the scikit-learn machine learning library. This synthetic dataset contains two interleaving crescent-shaped clusters that represent two classes.

The dataset was chosen because it is not linearly separable, making it suitable for demonstrating the limitations of linear models and the advantages of non-linear kernel methods. All models were implemented using the Support Vector Classifier (SVC) provided by scikit-learn. The experiments compare the performance of different kernel functions, including linear, polynomial, and radial basis function (RBF), on the same dataset.

The analysis was implemented using Python and several scientific computing libraries. The main tools used in the project include:

- NumPy for numerical operations
- Matplotlib for visualisation
- Seaborn for heatmap plotting.
- Scikit-learn for machine learning models.

RESULTS AND VISUALISATION:

1. Dataset Distribution:

Figure 1 shows how the two classes from the make_moons dataset are distributed. The data points create two curved clusters that overlap, making it impossible to separate them with a straight line.
This pattern shows why classification algorithms that can handle non-linear relationships are needed.

2. Linear Kernel Decision Boundary:

The decision boundary produced by the linear SVM model is shown in **Figure 2**. Because the linear kernel restricts the classifier to a straight boundary, it is unable to follow the curved structure of the dataset. As a result, several data points are incorrectly classified.

Figure 1: illustrates the distribution of the synthetic dataset generated using the make_moons function.

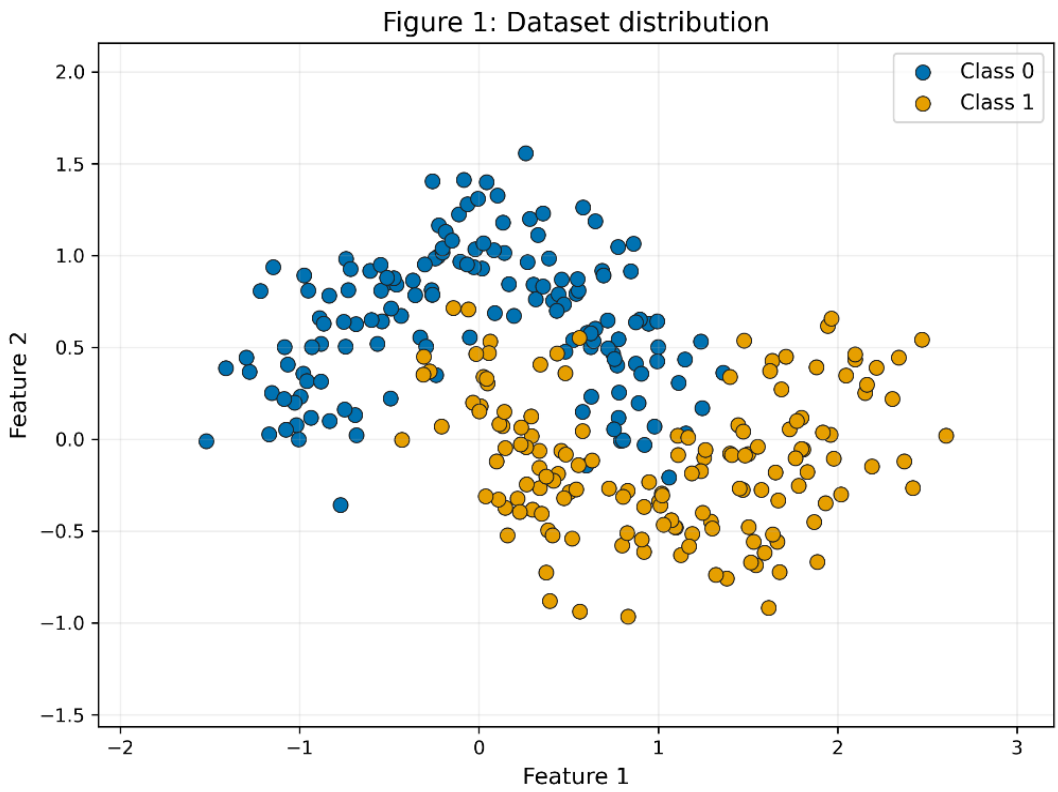
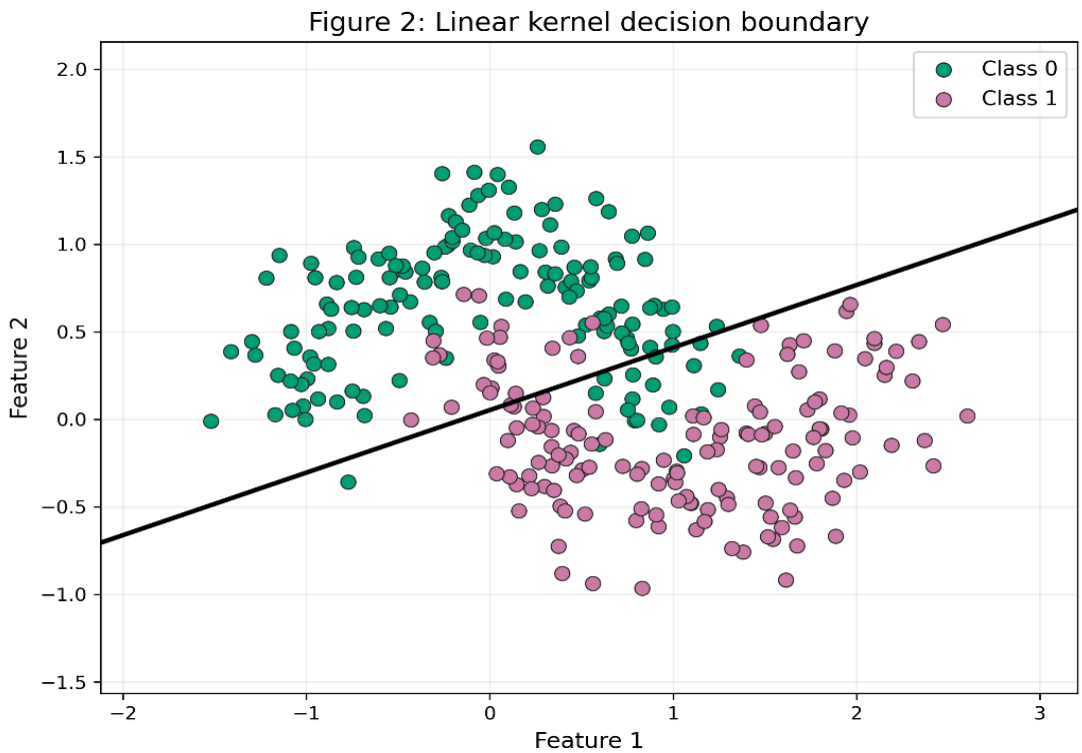


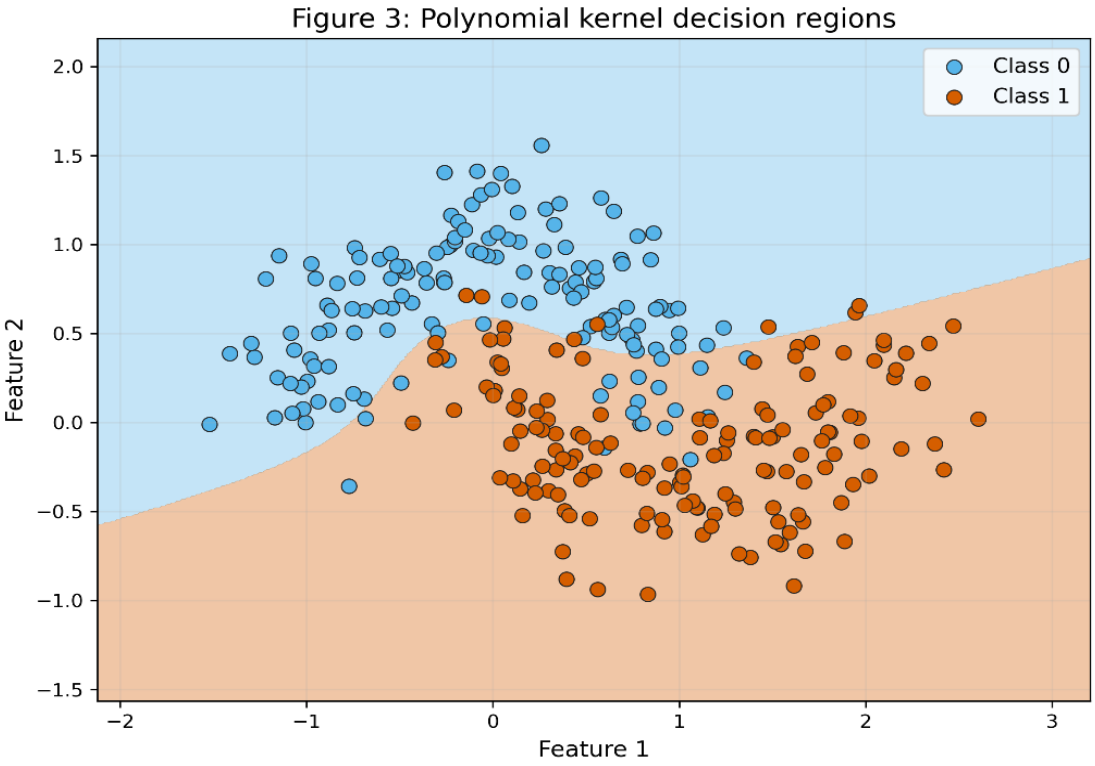
Figure 2: Decision boundary produced by a Support Vector Machine with a linear kernel. The straight boundary does not capture the curved structure of the dataset.



3. Polynomial Kernel Decision Regions:

Figure 3 explains the decision regions generated by the polynomial kernel. Compared with the linear model, the polynomial kernel introduces curvature into the decision boundary, enabling the classifier to better adapt to the structure of the dataset.

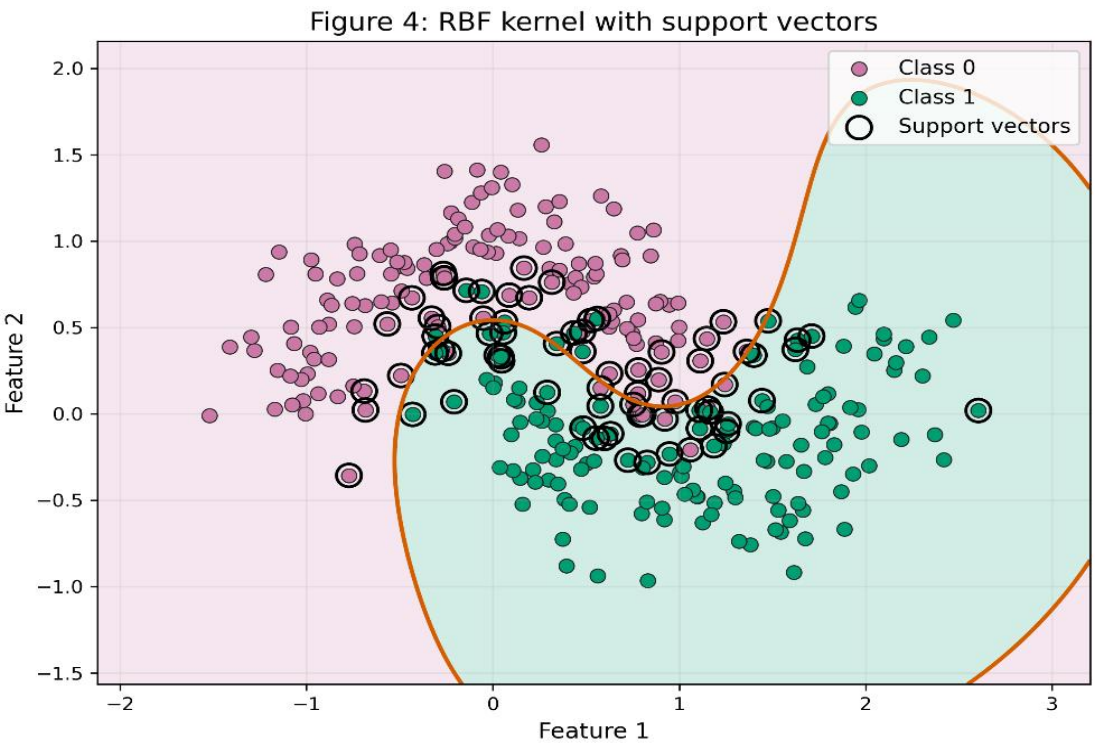
Figure 3: Decision regions produced by the polynomial kernel SVM. The curved boundary provides a better separation of the classes than the linear model.



4. RBF Kernel with Support Vectors:

The decision boundary produced by the RBF kernel is shown in **Figure 4**. In this figure, the support vectors are highlighted to demonstrate which data points have the greatest influence on the position of the decision boundary. The RBF kernel produces a highly flexible boundary that closely follows the dataset structure.

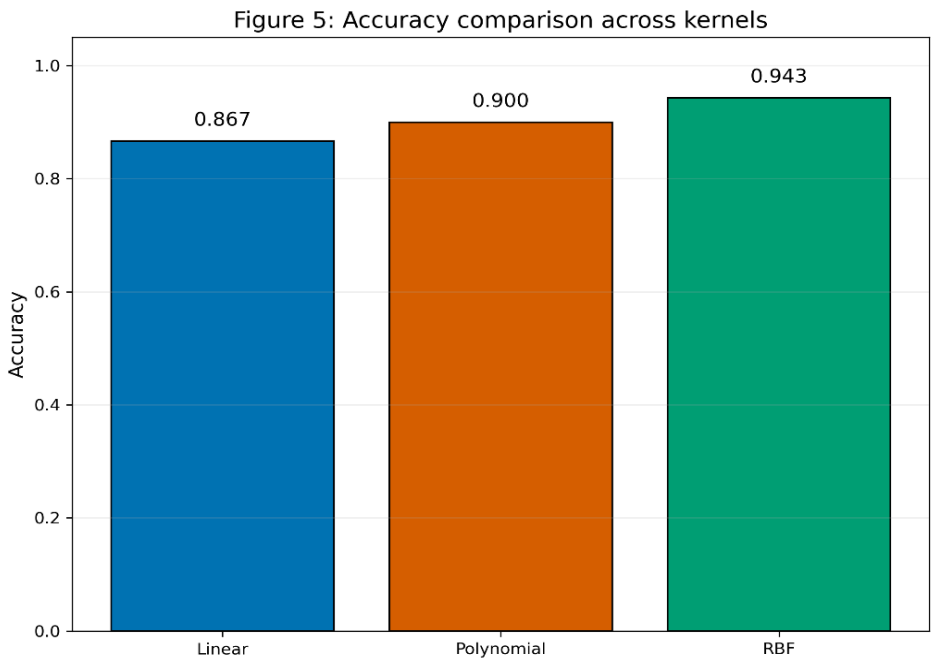
Figure 4: Decision boundary produced by the RBF kernel SVM. The highlighted support vectors represent the training points that define the classification boundary.



5. Kernel Accuracy Comparison:

A quantitative comparison of the kernel models is shown in **Figure 5**, which presents a bar chart of the classification accuracy achieved by each kernel. The results indicate that non-linear kernels outperform the linear kernel when applied to datasets with complex structures.

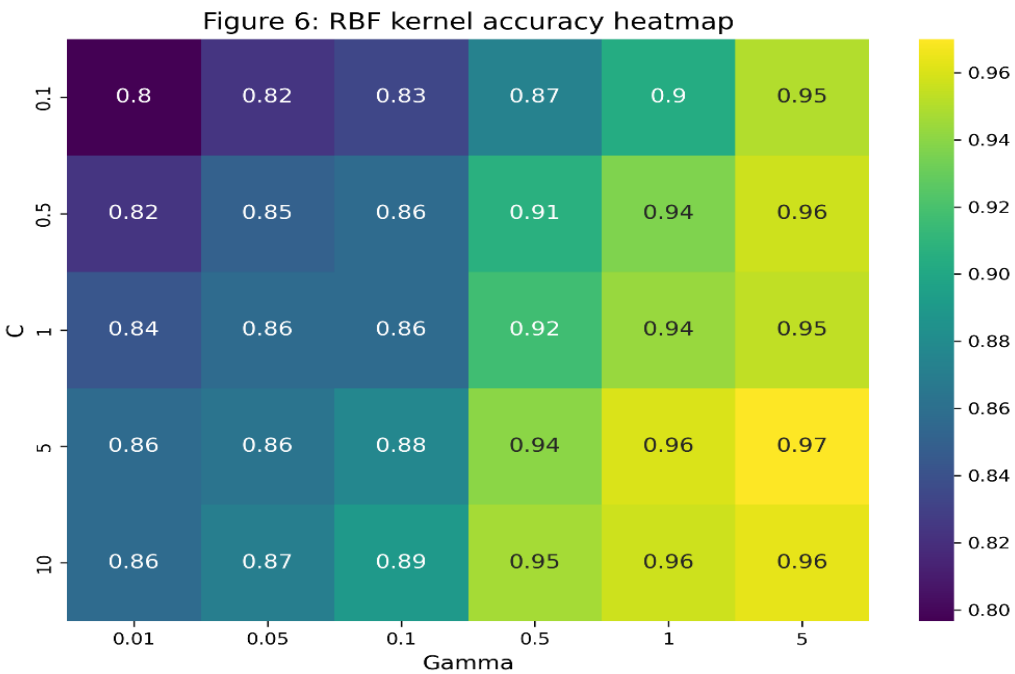
Figure 5: Bar chart comparing classification accuracy across linear, polynomial and RBF kernels.



6. Hyperparameter Heatmap:

Figure 6 shows how changing certain settings affects how well the model works. The chart uses colors to show how accurate the model is when using different values for C and gamma in the RBF kernel. Some combinations of these settings make the model much more accurate.

Figure 6: Heatmap showing classification accuracy of the RBF kernel for different combinations of the hyperparameters C and gamma.



DISCUSSION:

The linear kernel produces a simple dividing line that cannot separate the curved dataset well. On the other hand, the polynomial and RBF kernels are more flexible because they can oversee more complex patterns in the data. The results show that selecting different kernels has a substantial impact on how well Support Vector Machine classifiers work.

Figure 5 shows that the RBF kernel had the maximum classification accuracy at 0.943. The polynomial kernel followed with 0.900, and the linear kernel had the lowest at 0.867. These results suggest that non-linear kernels perform better for datasets with curved structures. This matches earlier research, which found that the RBF kernel performs well in many classification tasks (James et al., 2023). The hyperparameter heatmap also shows that tuning parameters like C and gamma can lead to different classification results.

CONCLUSION:

This report investigated how different kernel functions influence the behaviour of Support Vector Machine classifiers. Using a synthetic dataset with a non-linear structure, the analysis compared linear, polynomial, and radial basis function kernels.

The experiment results show that the type of kernel used is important for how well the SVM works. Of all the models tested, the RBF kernel gave the best results for the non-linear dataset. The study also found that picking the right settings for the model is important for how well it works.
Overall, the findings show that choosing the right kernel and settings is important when using Support Vector Machines for difficult classification problems.

ACCESSIBILITY CONSIDERATION:

The visualisations in this study were designed using colour-blind-friendly colour palettes and high-contrast markers to ensure accessibility. Clear labels and consistent formatting were used to make the figures easy to interpret.

REFERENCES:

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